

Prepared By	08-Aug-2012	Mr.Chatchawan S.	 Renew coils Engineering & Supply Co., Ltd.	
Checked By	08-Aug-2012	Mr.Chatchawan S.		
Approved By	08-Aug-2012	Mr.Amnauy W.		
Certified By	08-Aug-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	08-Aug-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-016-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-003**



RENEW COILS ENGINEER & SUPPLY CO., LTD.

WELDING PROCEDURE SPECIFICATION (WPS)

WELDING PROCEDURE SPECIFICATION (WPS) Yes ☒
PREQUALIFIED ☒ QUALIFIED BY TESTING ☐
or PROCEDURE QUALIFICATION RECORD (PQR) ☐

Company Name Renew coils and Engineering & Supply co., ltd.
Welding Process (es) SMAW
Supporting PQR No.(s) RNC-PQR-003

Identification # RNC-WPS-003
Revision 0 Date 8-Aug-12 By Mr. Chatchawan s.
Authorized by Mr. Amnuy V. Date 8-Aug-12
Type-Manual ☒ Semi-Automatic ☐
Machine ☐ Automatic ☐

JOINT DESIGN USED

Type: Butt Weld
Single ☒ Double Weld ☐
Backing: Yes ☐ No ☒
Backing Material: NONE
Root Opening 3 Root Face Dimension 3
Groove Angle: 60°±5 Radius (J-U) NONE
Back Gouging: Yes ☒ No ☐ Method

BASE METALS

Material Spec. SS400 / A36 or Strength Equivalence
Type or Grade Group 1 Carbon
Thickness: Groove 3.0 mm - Unlimited Fillet N/A
Diameter (Pipe)

FILLER METALS

AWS Specification A 5.1
AWS Classification E7016
Brand Name KOBELCO LB 52

SHIELDING

Flux N/A Gas N/A
Composition
Electrode-Flux (Class) N/A Flow Rate
Gas Cup Size

PREHEAT

Preheat Temp., Min 30 C°
Interpass Temp., Min 100C° Max 250 C°

POSITION

Position of Groove: 3G Fillet: None
Vertical Progression: Up ☒ Down ☐

ELECTRICAL CHARACTERISTICS

Transfer Mode (GMAW) Short-Circuiting ☐
Globular ☐ Spray ☐
Current: AC ☒ DCEP ☐ DCEN ☐ Pulsed ☐
Other
Tungsten Electrode (GTAW)
Size: NONE
Type: NONE

TECHNIQUE

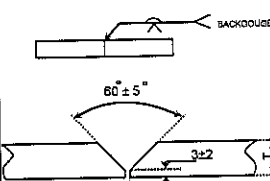
Stringer or Weave Bead: Stringer and Weave Bead
Multi-pass or Single Pass (per side) Multi-pass
Number of Electrodes
Electrode Spacing Longitudinal N/A
Lateral N/A
Angle N/A

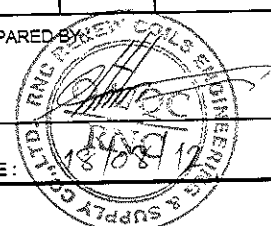
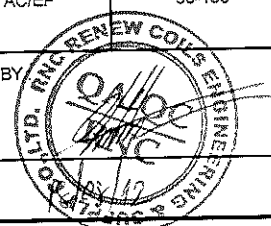
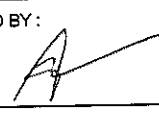
Contact Tube to Work Distance
Peening N/A
Interpass Cleaning: Grinding or Wire Brush

POSTWELD HEAT TREATMENT

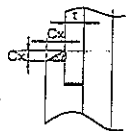
Temp. N/A
Time N/A

WELDING PROCEDURE

Pass or Weld Layer(s)	Process	Filler Metals		Type & Polarity	Current (Amps) or Wire Feed Speed	Volts	Travel Speed cm/min	Joint Details
		Class	Diam.					
Root pass	SMAW	E7016	3.2	AC/EP	80-120	20-25	5-12	
Hot pass	SMAW	E7016	3.2	AC/EP	80-120	20-25	5-12	
Filling pass	SMAW	E7016	4	AC/EP	130-180	25-28	7-15	
cap	SMAW	E7016	3.2	AC/EP	90-130	20-25	5-12	

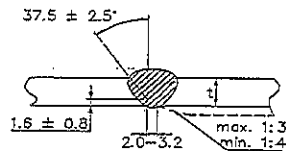
PREPARED BY: REVIEWED BY: APPROVED BY: DATE: 18/08/12DATE: 20/08/12DATE: 20-08-12

ATTACHED FIGURE JOINT SKETCH



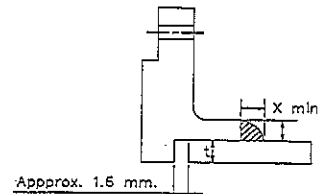
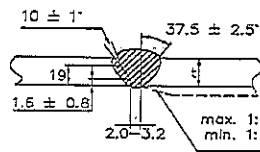
C_x (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



Misalignment (max.) = 1.5 mm.

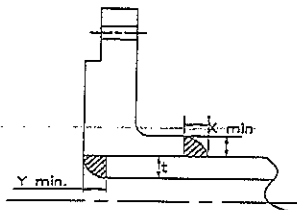
JOINT No. B1 & B2



Approx. 1.6 mm.

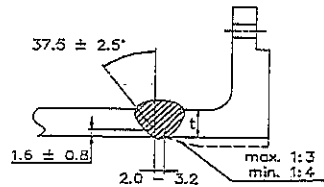
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



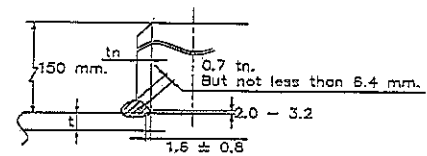
X min. = The lesser of $1.4t$ or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



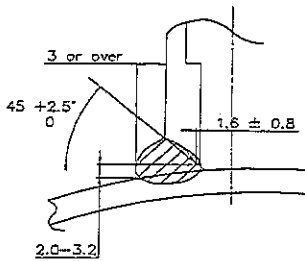
Misalignment (max.) = 1.5 mm.

JOINT No. B3

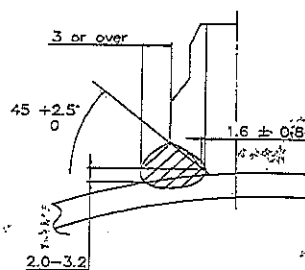


Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

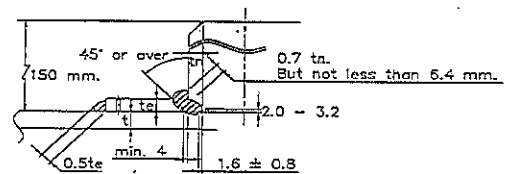
JOINT No. B4



JOINT No. C1

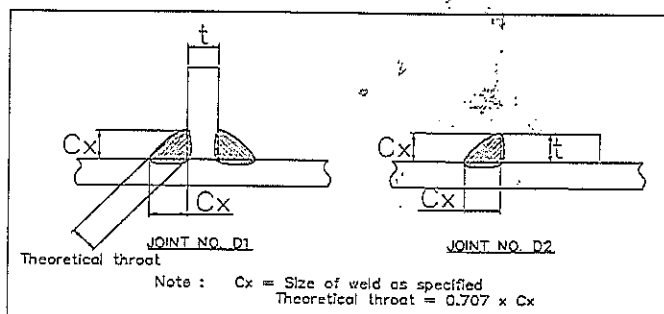


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

JOINT No. B5



[illegible]



PAE TECHNICAL SERVICE COMPANY LIMITED
69 Soi On-nuch 64, Srinakarin Road, Suanluang, Suanluang, Bangkok 10250
Tel. : (662) 322-0222 (Auto 20 lines) Fax. : (662) 721-2577

PQR No. : RNC-PQR-003

PROCEDURE QUALIFICATION RECORD (PQR)

NDT and Mechanical Test Result

TENSILE TEST (Report No : TS-12-08-081)						
Specimen No.	Width mm.	Thickness mm.	Area	Ultimate Load (KN)	Ultimate Strength (N/mm. ²)	Character of Failure and Location
3180/12 TR1	19.99	24.72	494.153	215.550	436.201	OUT OF WELD
3180/12 TR2	19.90	24.86	494.714	215.942	436.499	OUT OF WELD

GUIDE BEND TEST (Report No : BD-12-08-020)			
Specimen No	Type of Bend	Test Result	Location of Indication
3180/12 BS1	Transverse Side Bend	No Open Discontinuity	N/A
3180/12 BS2	Transverse Side Bend	No Open Discontinuity	N/A
3180/12 BS3	Transverse Side Bend	No Open Discontinuity	N/A
3180/12 BS4	Transverse Side Bend	No Open Discontinuity	N/A

Visual Inspection, PQR Report No:

PQR-010

Radiographic Examination

Appearance: Acceptable
Undercut/Slags: No Indication
Surface Porosity: No Indication
Test Date: 28 Jul 12
Witness By: Mr.Wichit K.
Organization: PAE Technical Service Co.,Ltd.

RT Report No : RT-PQR-015 Result : Accepted
Organization : PAE Technical Service Co.,Ltd.

Other Tests

1) Vicker Hardness Test

Report No: HN-12-08-033

3) Fillet Weld Test

Report No: N.A.

2) Macrostructure Examination

Report No: MA-12-08-024

4) Impact Test

Report No: N/A

Welder's Name : Mr.Choochat M.

Welder No : -

Mechanical Test Conducted By : LPN Plate Mill Public Company Limited

Test Conducted By : Mr.Keerati Sa-ngoen

Report No : -

We, the undersigned, certify the statements in this record are correct and that the test welds were prepared, welded, and Tested in conformance with the requirements of Clause Section 4 of AWS D1.1 Structural Welding Code-Steel.,

Manufacturer by
State Construction Co.,Ltd.

Signature:

Name

Date

19/08/12

Contractor by
PAE Technical Service Co.,Ltd.

Signature:

Name

Date

19/08/12



PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

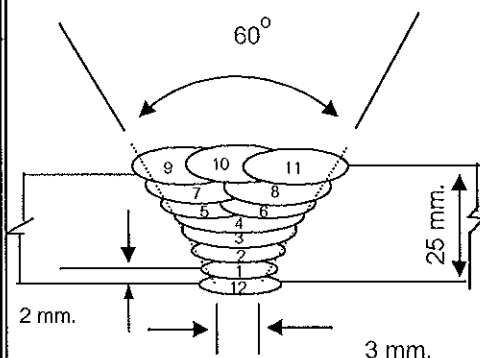
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 2

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR			Location : RNC Work Shop		Report No. PQR-010		
		Customer Order No. : N/A			WPS No. : RNC-WPS-003		Date : 6-Aug-12		
Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.CHOCHAT M.	SMAW	3G	AWS D1.1	JIS G3101 Gr.SS400	25x290x380	A 5.1	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	SMAW	E 7016	3.2	DCEP	127	24	5.02	35	36.43	-	-	-	-
2	SMAW	E 7016	3.2	DCEP	132	25	7.33	123	27.02	-	-	-	-
3	SMAW	E 7016	3.2	DCEP	131	25	5.38	131	36.53	-	-	-	-
4	SMAW	E 7016	3.2	DCEP	132	25	5.17	172	38.30	-	-	-	-
5	SMAW	E 7016	3.2	DCEP	135	25	9.39	225	21.57	-	-	-	-
6	SMAW	E 7016	3.2	DCEP	132	25	8.68	182	22.82	-	-	-	-
7	SMAW	E 7016	3.2	DCEP	137	25	8.62	230	23.84	-	-	-	-
8	SMAW	E 7016	3.2	DCEP	132	25	11.05	236	17.92	-	-	-	-
9	SMAW	E 7016	3.2	DCEP	134	25	8.60	223	23.38	-	-	-	-
10	SMAW	E 7016	3.2	DCEP	132	25	10.74	198	18.44	-	-	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

Mr.Wichit

7-Aug-2012



SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :





PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

WELDER PROCEDURE QUALIFICATION RECORD

Page No. 2 of 2

Client : Renew Coils Engineering & Supply Co.,Ltd.		Project Name : PQR		Location : RNC Work Shop		Report No. PQR-010	
		Customer Order No. : N/A		WPS No. : RNC-WPS-003		Date : 6-Aug-12	

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.CHOOCHAT M.	SMAW	3G	AWS D1.1	JIS G3101 Gr.SS400	25x290x380	A 5.1	ACCEPTED	ACCEPTED

Joint Details

Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
11	SMAW	E 7016	3.2	DCEP	134	25	9.05	126	22.21	-	-	-	-
12	SMAW	E 7016	3.2	DCEP	128	24	12.11	100	15.22	-	-	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

Mr.Wichit K

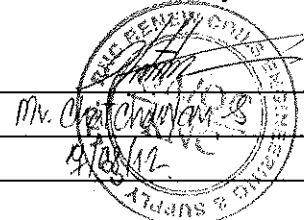
7-Aug-2012



SIGNATURE :

NAME :

DATE :



SIGNATURE :

NAME :

DATE :



[illegible]



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedgee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th



23853

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-12-08-081

Revision No. 0

Test Product	PQR No. RNC-PQR-003	Material Characterization	Plate
Project Name	N/A	Test Method	AWS D1.1/D1.1 M:2010
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400 / A36	Type of Test	Reduced Section Tension
Dimension	25 mm Thk.	Equipment's Capacity	200 TONS
Date Received	10 August 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	14 August 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	3180/12 TR1	N/A	24.72	19.99	494.153	N/A	N/A	N/A	215.550	436.201	N/A	OUT OF WELD
2	3180/12 TR2	N/A	24.86	19.90	494.714	N/A	N/A	N/A	215.942	436.499	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-003

Welding Process : SMAW

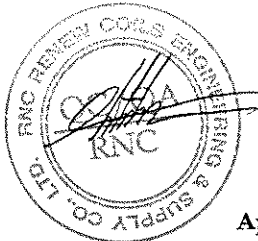
Filler Metal/Electrode Spec't : E7016

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 16 / 08 / 12



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 16 / 08 / 12

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.

3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th



23858

TESTING
No. 0154

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-12-08-020

Revision No. 0

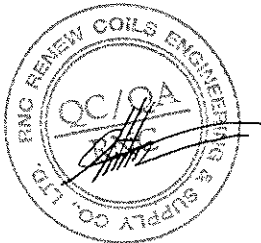
Test Product	PQR No. RNC-PQR-003	Material Characterization	Plate
Project Name	N/A	Test Method	AWS D1.1/D1.1 M:2010
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400 / A36	Type of Jig	Guided-Bend Roller Jig
Dimension	25 mm Thk.	Diameter of Mandrel	38.1 mm
Date Received	10 August 2012	Bend Angle	180 Degree.
Date Tested	14 August 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	3180/12 BS1	Transverse side bend	9.81x23.75x162.75 mm	No Open Discontinuity	N/A	-
2	3180/12 BS2	Transverse side bend	9.86x23.73x162.96 mm	No Open Discontinuity	N/A	-
3	3180/12 BS3	Transverse side bend	9.89x23.81x162.59 mm	No Open Discontinuity	N/A	-
4	3180/12 BS4	Transverse side bend	9.98x23.83x163.29 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-003

Welding Process : SMAW

Filler Metal/Electrode Spec't : E7016



Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 16 / 08 / 12

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 16 / 08 / 12

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.

3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

12747

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-08-033

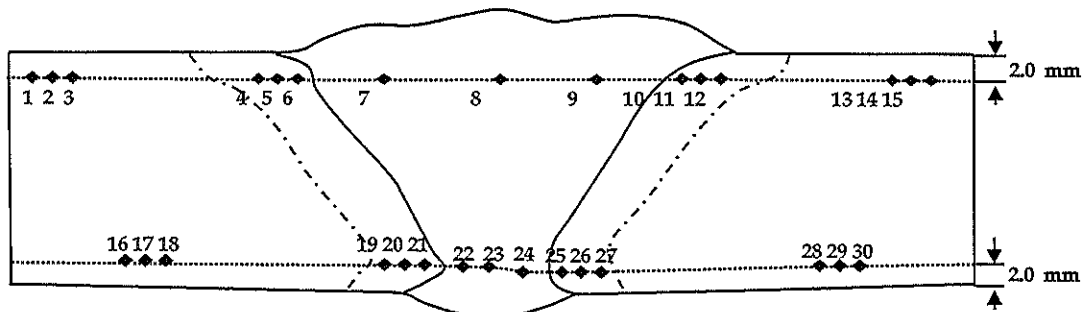
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-003	Material Characterization	Plate
Test No.	3180/12 HN	Test Method	ASTM E92-03
Project Name	N/A	Standard of Test	AWS D1.1/D1.1M : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	JIS G3101 SS400	Equipment	Mitutoyo, HV-115
Dimension	25 mm Thk.	Indenter Size	Diamond, pyramid 136 deg.
Welding Process/Position	SMAW/3G	Test Load	HV 10 Kgf.
Type of Joint	Butt joint	Total Force Dwell Time	15 s
Date Received	10 August 2012	Cal. Block	No. 62137 No. A62072
Date Tested	14 August 2012	Ambient Temperature	25.0 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	133.4	16	BASE	142.8	*****		
2	BASE	134.0	17	BASE	142.5			
3	BASE	134.6	18	BASE	141.2			
4	HAZ	169.1	19	HAZ	153.3			
5	HAZ	170.8	20	HAZ	157.0			
6	HAZ	178.4	21	HAZ	167.0			
7	WELD	193.1	22	WELD	208.8			
8	WELD	185.9	23	WELD	204.3			
9	WELD	177.8	24	WELD	205.6			
10	HAZ	165.8	25	HAZ	174.3			
11	HAZ	164.0	26	HAZ	161.1			
12	HAZ	162.6	27	HAZ	153.6			
13	BASE	130.3	28	BASE	141.4			
14	BASE	131.6	29	BASE	140.2			
15	BASE	130.1	30	BASE	140.4			

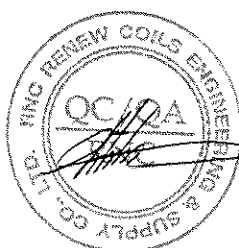
Additional details: Filler Metal : E7016(SMAW)

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurancer

Date : 15/08/12

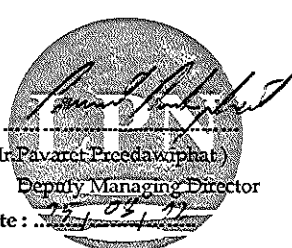


Approved by :

(Mr.Pavarat Freedawiphat)

Deputy Managing Director

Date : 15/08/12



LPN PLATE MILL

Notes

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



Macrostructure Analysis Report

Page 1 of 1

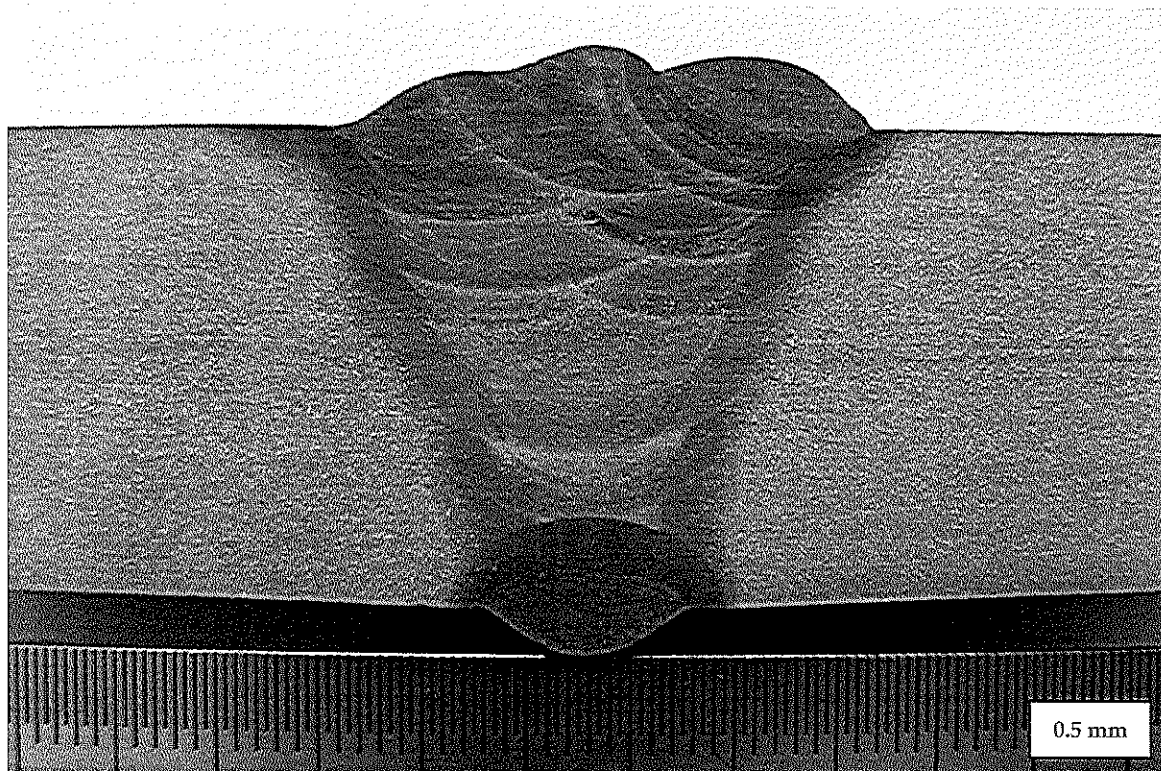
Customer : RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

Report No. MA-12-08-024

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-003	Material Characterization	Plate
Test No.	3180/12 MA	Test Method	ASME E340-00
Project Name	N/A	Standard of Test	AWS D1.1/D1.1M : 2010
Project No.	N/A	Etchant Reagent	75ml H ₂ O+25ml HNO ₃ (65%)
Material Specification	JIS G3103 SS400	Temperature	28 °C
Dimension	25 mm Thk.	Time of Etching	10 minutes
Welding Process/Position	SMAW/3G	Surface Preparation	Polishing with sand paper 240, 400, 600, 800, 1000
Type of Joint	Butt joint	Date Tested	14 August 2012
Date Received	10 August 2012		



Additional details : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

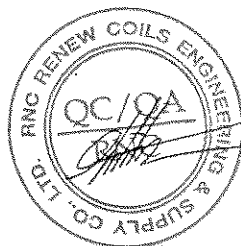
Filler Metal : E7016(SMAW)

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurancer

Date : 15/08/12



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 15/08/12



LPN PLATE MILL

Notes

1. The above results are valid exclusively for tested samples as mentioned in this report.
2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.
3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnrm.co.th



3863

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-08-086

Revision No. 0

Test Product	PQR No. RNC-PQR-003	Material Characterization	Plate
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ¹
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400 / A36	Type of Test	Charpy V-Notch Impact
Dimension	25 mm Thk.	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	10 August 2012	Equipment 's Capacity	542 Joules
Date Tested	15 August 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	3180/12-1	-20°C	10.018	10.013	54.95	153.27	157.41	Weld Metal	-
2	3180/12-2	-20°C	10.017	10.006	54.95	133.20			
3	3180/12-3	-20°C	10.016	10.011	54.93	185.76			
4	3180/12-1	-20°C	10.011	10.016	54.99	299.37	276.23	Fusion Line+1mm	-
5	3180/12-2	-20°C	10.018	10.011	54.99	277.03			
6	3180/12-3	-20°C	10.016	10.005	54.99	252.28			
7	3180/12-1	-20°C	10.016	10.008	54.96	12.09	10.72	Fusion Line+5mm	-
8	3180/12-2	-20°C	10.012	10.015	54.98	10.08			
9	3180/12-3	-20°C	10.014	10.010	54.92	9.99			

Additional details : WPS No. RNC-WPS-003

Welding Process : SMAW

Filler Metal/Electrode Spec't : E7016

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

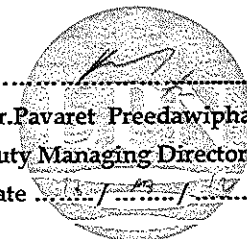
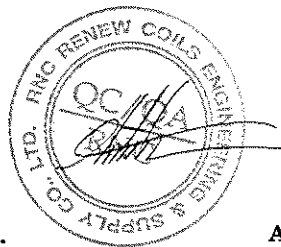
Date/...../.....

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date/...../.....

**Notes** 1. The above results are valid exclusively for tested samples as mentioned in this report.

2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.

3. This report shall not be reproduced in whole or in part without the written approval of LPN LABORATORY.

LPN PLATE MILL

24/11/54

KOBELCO

INSPECTION CERTIFICATE

PURCHASER

Date of Issue 11/Aug/2011

Certificate No. TF11-035

Date of Inspection 16/Aug/2011

TRADE DESIGNATION	SIZE (mm)	MFG. No.	APPLICABLE SPECIFICATION & CLASSIFICATION								
LB-52	3.2 x 350	12110	AWS A5.1 E7016								
Chemical Composition (%)											
Elements	C	Si	Mn	P	S	Ni	Cr	Mo	V	*1*	
Weld Metal	0.07	0.52	0.93	0.009	0.003	< 0.01	0.03	< 0.01	0.009	0.97	
Tensile Test of Weld Metal						Impact Test of Weld Metal					
Yield Strength at 0.2% offset	Tensile Strength		Elongation		Test Temp.	Absorbed Energy		Tested size for Mechanical test: 4.0x400 mm Tested MFG. No. of Mechanical test: 12009			
465 MPa	554 MPa		33 %		-30 °C	Avg.	246 J				
Welding Conditions						Postweld Heat Treatment			REMARKS		
Type of Current	AC					--- °C x --- Hr.			*1* : Mn+Ni+Cr+Mo+V		
Welding Current	120 A										
Welding Voltage	24 V										

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.



THAI-KOBE WELDING CO., LTD.

CHIEF INSPECTOR

[Signature]



